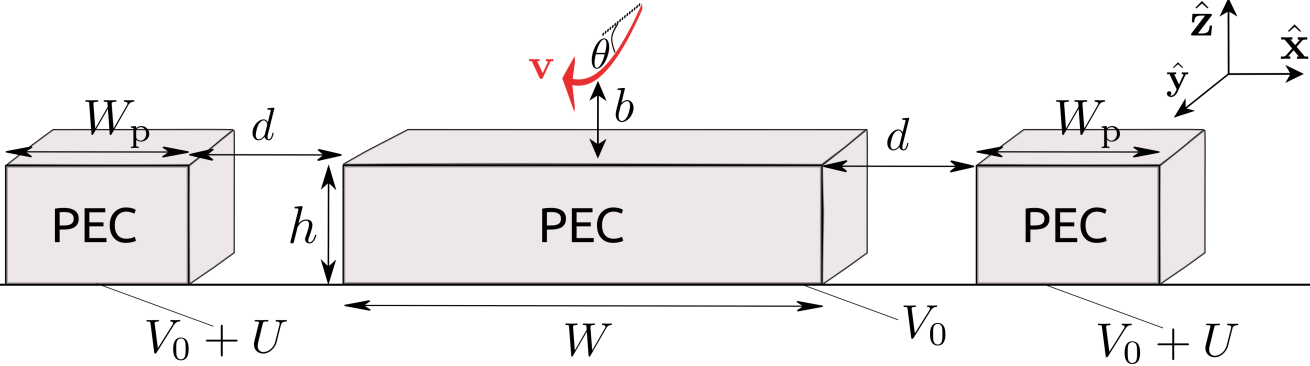


Electron-coupling to waveguide

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{b_{\text{min}}+h}^{\infty} \frac{v dz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x,z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy}$$

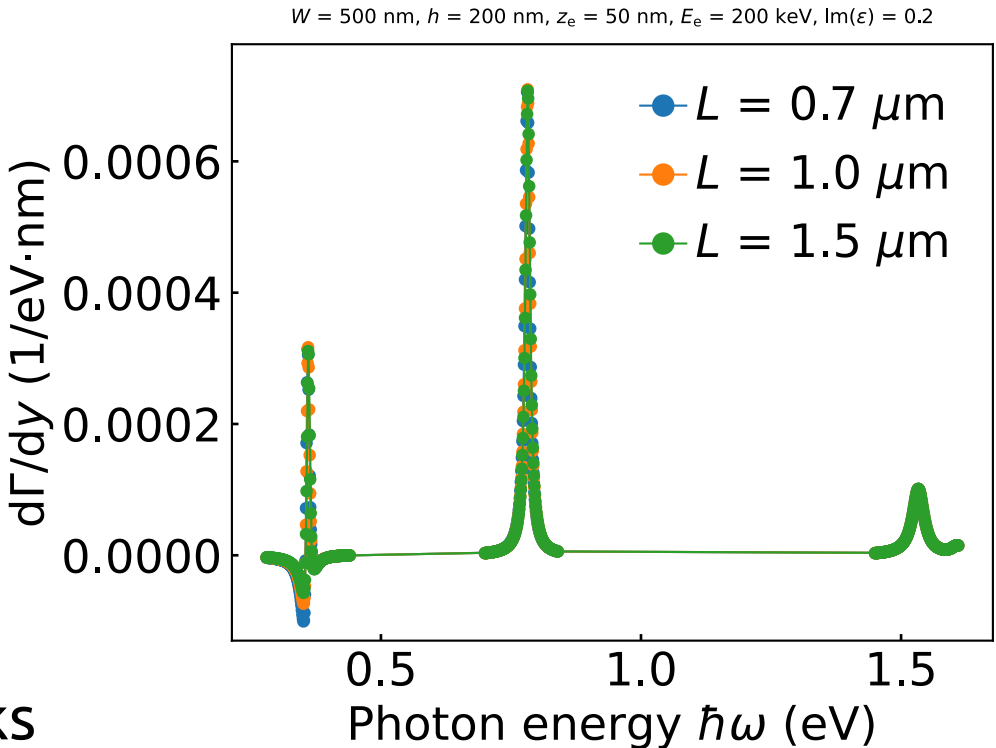
$z=0$ $\epsilon=9$

from BEM2D



$E_e = 200 \text{ keV}$ $W_p/W = 0.5$ $d/W = 0.2$ $x/W = 0$ $h/W = 0.4$ $W = 500 \text{ nm}$ $\epsilon_{\text{Si}} = 12.25 + i0.2$

Convergency for L = 1000 nm
(need Fadil's cores in sonic)



Higher resolution around the peaks